

20.5 Masking and swamping

Every diagnostic so far removes *one* case. But bad cases often come in groups: a batch of records in the wrong units, or a sensor that failed for an afternoon. **Masking** is when a bad case escapes detection because others hide it: deleting it alone leaves its companions to hold the fit in place. **Swamping** is when good cases are flagged because the bad ones have dragged the fit away from them. This section deletes sets of cases and proves how a cluster masks itself and swamps the rest.

20.5.1 Deleting a set of cases

Let $D \subset \{1, \dots, n\}$ be a set of d cases. Write $\mathbf{X}_D$ for the $d \times p$ matrix of their rows, $\mathbf{X}_{(D)}$ for $\mathbf{X}$ with those rows removed, $\hat{\mathbf{e}}_D$ for their residuals in the full fit, and

$$\mathbf{H}_D = \mathbf{X}_D(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top$$

for the $d \times d$ block of the hat matrix that belongs to them. Let $\hat{\boldsymbol{\beta}}_{(D)}$ and $\text{SSE}_{(D)}$ come from the least squares fit to the cases not in D , and let $\mathbf{E}_D$ be the $n \times d$ matrix whose columns are the coordinate vectors $\mathbf{e}_i, i \in D$.

Theorem 20.5.1. Deleting a set of cases

Let $\mathbf{X}$ and $\mathbf{X}_{(D)}$ have full column rank. Then $\mathbf{I} - \mathbf{H}_D$ is nonsingular, and

- (a) $\mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$;
- (b) $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$;
- (c) $\text{SSE}_{(D)} = \text{SSE} - \hat{\mathbf{e}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D$;
- (d) if $n - p - d \geq 1$, then under Eq. 7.3.1

$$F_D = \frac{\hat{\mathbf{e}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D / d}{\text{SSE}_{(D)} / (n - p - d)} \sim F(d, n - p - d),$$

and F_D is the F statistic for $\boldsymbol{\gamma} = \mathbf{0}$ in the mean-shift model $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{E}_D\boldsymbol{\gamma}$.

Proof. Let $\mathbf{Z} = [\mathbf{X}, \mathbf{E}_D]$. If $\mathbf{X}\mathbf{b} + \mathbf{E}_D\mathbf{c} = \mathbf{0}$, the rows outside D give $\mathbf{X}_{(D)}\mathbf{b} = \mathbf{0}$, so $\mathbf{b} = \mathbf{0}$, and then $\mathbf{c} = \mathbf{0}$. So $\mathbf{Z}$ has full column rank. By Lemma 6.6.1(b) with $\mathbf{X}_1 = \mathbf{E}_D$, the matrix $(\mathbf{I} - \mathbf{M})\mathbf{E}_D$ has full column rank, so its Gram matrix $\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D = \mathbf{I} - \mathbf{H}_D$ is nonsingular.

Fit the mean-shift model. By Theorem 6.6.2(a) with $\mathbf{X}_1 = \mathbf{E}_D$ and $\mathbf{X}_2 = \mathbf{X}$, the estimated shifts are

$$\hat{\boldsymbol{\gamma}} = (\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D.$$

On the other hand, for any $\mathbf{b}$ the residual sum of squares $\sum_{i \notin D} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + \|\mathbf{y}_D - \mathbf{X}_D \mathbf{b} - \mathbf{c}\|^2$ is minimized over $\mathbf{c}$ by $\mathbf{c} = \mathbf{y}_D - \mathbf{X}_D \mathbf{b}$, which removes the second term. What remains is minimized by $\mathbf{b} = \hat{\boldsymbol{\beta}}_{(D)}$, uniquely because $\mathbf{X}_{(D)}$ has full column rank. So the mean-shift fit has coefficients $\hat{\boldsymbol{\beta}}_{(D)}$ and $\hat{\boldsymbol{\gamma}} = \mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)}$, and its residual sum of squares is $\text{SSE}_{(D)}$. Comparing the two expressions for $\hat{\boldsymbol{\gamma}}$ gives (a). By Theorem 6.6.2(c), the coefficients of $\mathbf{X}$ in the mean-shift fit are $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{E}_D \hat{\boldsymbol{\gamma}}) = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \hat{\boldsymbol{\gamma}}$, which gives (b). By Lemma 6.6.1(c) the projection onto $\mathcal{C}(\mathbf{Z})$ is $\mathbf{M}$ plus the projection onto $\mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{E}_D)$, so

the fall in the residual sum of squares from the fit of $\mathbf{X}$ to the fit of $\mathbf{Z}$ is

$$\mathbf{y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{E}_D (\mathbf{I} - \mathbf{H}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M}) \mathbf{y} = \hat{\boldsymbol{\varepsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\varepsilon}}_D,$$

which gives (c). Finally $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ with dimensions p and $p + d$, and F_D is the statistic Eq. 11.1.1 for this pair of nested models, so (d) is Theorem 11.1.1(d). ■

With $d = 1$ this is Theorem 20.3.1, and $F_D = t_i^2$ (Exercise (A2)). Part (d) is exact only for a set chosen in advance; a Bonferroni correction over all $\binom{n}{d}$ subsets is valid but very conservative.

20.5.2 How a cluster masks itself

The simplest mechanism is a group of identical bad cases: n_0 clean cases plus m copies of one bad case.

Proposition 20.5.2. Masking and swamping by a cluster

Let $\mathbf{X}_0$ ($n_0 \times p$, full column rank) and $\mathbf{y}_0$ be the clean cases, with least squares estimate $\hat{\boldsymbol{\beta}}_0$, residuals $\tilde{\varepsilon}_j = y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}}_0$ and residual sum of squares SSE_0 . Add $m \geq 1$ cases, each with row $\mathbf{x}_\star^\top$ and response $y_\star$. Put $\mathbf{A} = \mathbf{X}_0^\top \mathbf{X}_0$,

$$h_\star = \mathbf{x}_\star^\top \mathbf{A}^{-1} \mathbf{x}_\star, \quad \delta = y_\star - \mathbf{x}_\star^\top \hat{\boldsymbol{\beta}}_0,$$

the leverage of $\mathbf{x}_\star$ relative to the clean design and the distance of the cluster from the clean fit. In the least squares fit to all $n_0 + m$ cases:

- (a) each cluster case has leverage $h = h_\star / (1 + mh_\star) < 1/m$;
- (b) each cluster case has residual $\delta / (1 + mh_\star)$;
- (c) deleting one cluster case gives it prediction residual $\delta / (1 + (m-1)h_\star)$ and changes the estimate by $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{A}^{-1} \mathbf{x}_\star \delta / [(1 + mh_\star)(1 + (m-1)h_\star)]$;
- (d) deleting the whole cluster gives each member prediction residual δ , and $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = m(1 + (m-1)h_\star)(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})$;
- (e) clean case j has residual $\tilde{\varepsilon}_j - m\delta h_{j\star} / (1 + mh_\star)$, where $h_{j\star} = \mathbf{x}_{(j)}^\top \mathbf{A}^{-1} \mathbf{x}_\star$, and $\text{SSE} = \text{SSE}_0 + m\delta^2 / (1 + mh_\star)$.

Proof. Let D be the cluster. Then $\mathbf{X}^\top \mathbf{X} = \mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top$, and $(\mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top) \mathbf{A}^{-1} \mathbf{x}_\star = (1 + mh_\star) \mathbf{x}_\star$, so

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = \frac{\mathbf{A}^{-1} \mathbf{x}_\star}{1 + mh_\star}. \quad (20.5.1)$$

- (a) $h = \mathbf{x}_\star^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = h_\star / (1 + mh_\star)$, and $mh = mh_\star / (1 + mh_\star) < 1$.
- (b) and (d) Deleting D leaves the clean cases, so $\hat{\boldsymbol{\beta}}_{(D)} = \hat{\boldsymbol{\beta}}_0$ and every cluster case has prediction residual δ . All rows of $\mathbf{X}_D$ equal $\mathbf{x}_\star^\top$, so $\mathbf{H}_D = h \mathbf{1} \mathbf{1}^\top$, and by Theorem 20.5.1(a) $\hat{\boldsymbol{\varepsilon}}_D = (\mathbf{I} - h \mathbf{1} \mathbf{1}^\top) \delta \mathbf{1} = (1 - mh) \delta \mathbf{1} = \delta \mathbf{1} / (1 + mh_\star)$. By Theorem 20.5.1(b) and Eq. 20.5.1, $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \delta \mathbf{1} = m\delta (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = m\delta \mathbf{A}^{-1} \mathbf{x}_\star / (1 + mh_\star)$.

- (c) By Theorem 20.3.1(d) and (b), with $1 - h = (1 + (m - 1)h_*)/(1 + mh_*)$, the prediction residual is $\delta/(1 + (m - 1)h_*)$. By Theorem 20.3.1(b) and Eq. 20.5.1, $\hat{\beta} - \hat{\beta}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_* \cdot \delta/(1 + (m - 1)h_*)$, which is the stated vector. Comparing with the expression for $\hat{\beta} - \hat{\beta}_0$ in the previous step gives the ratio in (d).
- (d) $y_j - \mathbf{x}_{(j)}^\top \hat{\beta} = \tilde{\epsilon}_j - \mathbf{x}_{(j)}^\top (\hat{\beta} - \hat{\beta}_0)$, and $\mathbf{x}_{(j)}^\top (\hat{\beta} - \hat{\beta}_0) = m\delta h_{j*}/(1 + mh_*)$. For the sum of squares, Theorem 20.5.1(c) gives $\text{SSE} = \text{SSE}_0 + \hat{\epsilon}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\epsilon}_D$, and $\hat{\epsilon}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\epsilon}_D = \delta^2 \mathbf{1}^\top (\mathbf{I} - h\mathbf{1}\mathbf{1}^\top) \mathbf{1} = m\delta^2(1 - mh)$, using $(\mathbf{I} - h\mathbf{1}\mathbf{1}^\top)^{-1} \hat{\epsilon}_D = \delta \mathbf{1}$ from part (b).

■

Each part is a way of hiding. **Leverage is masked** (a): however remote the cluster, each copy has leverage below $1/m$; a single copy would have $h_*/(1 + h_*)$, near one. **Residuals are masked** (b): each is only $1/(1 + mh_*)$ of the cluster's distance from the clean fit. **Deletion is masked** (c, d): deleting one copy leaves $m - 1$ to hold the fit, recovering only $1/(1 + (m - 1)h_*)$ of δ . **Good cases are swamped** (e): clean residuals shift in proportion to their cross-leverage with the cluster, and s^2 is inflated.

Example. Four identical bad records

Thirty clean cases have x uniform on $(0, 10)$ and $y = 2 + 0.5x + \epsilon$, $\sigma = 0.5$. Four identical records are added at $x = 20$, $y = 2$, where the clean line predicts about 12 (Figure 20.5.1(a)). The slope drops from 0.514 (clean cases) to 0.007, and s rises from 0.608 to 1.632. Here $h_* = 1.126$ (it can exceed one, since $x = 20$ is outside the clean design, as in Proposition 12.3.2) and $\delta = -10.150$. Each bad case has leverage 0.205, below $1/m = 0.25$, where a single copy would have 0.530; residual -1.844 ; prediction residual -2.318 ; and $t_i = -1.28$. Deleting the group changes $\hat{\beta}$ 17.51 times as much as deleting one member.

The single-case diagnostics fail as predicted. The Bonferroni test (critical value 3.49) flags nothing, and the only case with $|t_i| > 2$ is a clean one: case 30, the clean case with largest x (9.86), has $t_i = 2.25$, because the dragged line passes below it. Cook's distance does a little better: each bad case has $D_i = 0.206$, above $4/n = 0.118$ (no clean case exceeds it; the largest is 0.100) but far below the median of $F(2, 32)$. A single bad record alone would have had $t_i = -11.45$ and $D_i = 13.46$. Deleting the four together tells the truth: their prediction residuals are all δ , and Theorem 20.5.1(d) gives $F = 50.6$ on 4 and 28 degrees of freedom, p -value 2.0×10^{-12} . The difficulty is not in testing the cluster but in finding it.

```
import numpy as np
from scipy import stats

def single_case(X, y):
    """Leverages, residuals, s^2, externally studentized residuals and Cook's
    distances."""
    n, p = X.shape
    Q, _ = np.linalg.qr(X)
    h = np.sum(Q ** 2, axis=1)
    e = y - Q @ (Q.T @ y)
    s2 = e @ e / (n - p)
    s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)
    t = e / np.sqrt(s2_del * (1 - h))
    cook = e ** 2 * h / (p * s2 * (1 - h) ** 2)
    return h, e, s2, t, cook
```

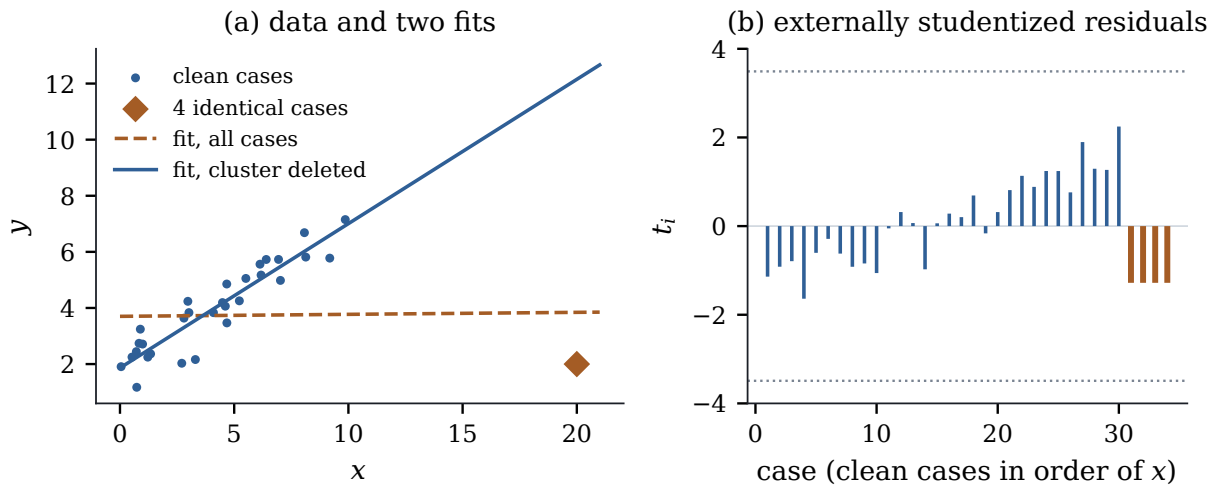


Figure 20.5.1: Masking and swamping. (a) Clean cases, four identical bad records (diamond), and least squares lines with and without them. (b) Externally studentized residuals (bad records last) and the 5% Bonferroni cut-offs.

```
rng = np.random.default_rng(20)
n0, m = 30, 4                                # clean cases, size of the
    cluster
x_clean = np.sort(rng.uniform(0, 10, n0))
y_clean = 2 + 0.5 * x_clean + rng.normal(0, 0.5, n0)
x0, y0 = 20.0, 2.0                            # four identical records
x = np.r_[x_clean, np.full(m, x0)]
y = np.r_[y_clean, np.full(m, y0)]
X = np.column_stack([np.ones(len(x)), x])
n, p = X.shape
cluster = np.arange(n0, n)

h, e, s2, t, cook = single_case(X, y)
crit = stats.t.isf(0.05 / (2 * n), n - p - 1)   # Bonferroni outlier test at 5%
print("slope with the cluster:", np.linalg.lstsq(X, y, rcond=None)[0][1])
print("cluster: h =", h[-1].round(3), " t =", t[-1].round(2), " Cook's D =", cook
      [-1].round(3))
i = int(np.argmax(np.abs(t)))
print(f"largest |t|: case {i + 1} at x = {x[i]:.2f}, t = {t[i]:.2f}; Bonferroni cut-
      off {crit:.2f}")
```

[Run this code online](#)

```
D = cluster                                # delete the whole cluster at
    once
K = np.linalg.inv(X.T @ X)
H_D = X[D] @ K @ X[D].T
pred_D = np.linalg.solve(np.eye(m) - H_D, e[D]) # y_D minus its prediction from
    the rest
sse = e @ e
sse_D = sse - e[D] @ pred_D
F = (e[D] @ pred_D / m) / (sse_D / (n - p - m))
print("group prediction residuals:", pred_D.round(3))
print(f"F = {F:.1f} on {m} and {n - p - m} degrees of freedom,"
      f" p-value {stats.f.sf(F, m, n - p - m):.1e}")
```

[Run this code online](#)

20.5.3 Finding the set

There are $\binom{n}{d}$ candidate sets of each size d , far too many to search. The natural shortcut, deleting the case with the largest $|t_i|$ and repeating, starts in the wrong place: in [Example \(Four identical bad records\)](#) it would first remove case 30, a clean case. Three ideas get around the combinatorics. First, **start from a fit the cluster cannot capture**, one determined by about half the data (the next section); the cases it flags form a candidate set to examine and test, remembering that it was chosen by looking. Second, **grow a clean subset**: the forward search of Atkinson and Riani (2000) adds cases one at a time in order of their residuals from a robust start, and a cluster enters late and all at once, as a jump in the monitored residuals and estimates; Hadi and Simonoff (1993) proposed a related procedure. Third, **unmask leverage separately**: by [Proposition 20.5.2\(a\)](#) the hat matrix is fooled by clusters, since leverage is a Mahalanobis distance built from the sample covariance, which a cluster inflates in its own direction. Rousseeuw and van Zomeren (1990) replaced the mean and covariance by estimates that ignore up to half the cases.

Idea. Single-case diagnostics answer “what if this case alone were different?”, the wrong question when bad cases come in groups. Group deletion asks the right one, but only for a suspected group; finding the group needs a fit it cannot capture.

20.5.4 Exercises

20.5.4.1 A. Check your understanding

Exercise. A1

Check the leverage 0.205 and the residual -1.844 in [Example \(Four identical bad records\)](#) from $h_\star = 1.126$, $\delta = -10.150$ and $m = 4$.

Exercise. A2

Show that [Theorem 20.5.1](#) with $D = \{i\}$ gives parts (b), (d) and (e) of [Theorem 20.3.1](#), and that $F_D = t_i^2$.

Solution. With $d = 1$, $\mathbf{H}_D = h_{ii}$ and $\hat{\boldsymbol{\epsilon}}_D = \hat{\epsilon}_i$, so (a) is the prediction residual $\hat{\epsilon}_i/(1 - h_{ii})$, (b) is [Theorem 20.3.1\(b\)](#), and (c) is $\text{SSE}_{(i)} = \text{SSE} - \hat{\epsilon}_i^2/(1 - h_{ii})$. Then $F_D = \hat{\epsilon}_i^2/\{(1 - h_{ii})s_{(i)}^2\} = t_i^2$.

20.5.4.2 B. Practice

Exercise. B1

Show that for $\mathbf{H}_D = h\mathbf{1}\mathbf{1}^\top$ ($d \times d$) with $dh \neq 1$, $(\mathbf{I} - h\mathbf{1}\mathbf{1}^\top)^{-1} = \mathbf{I} + h\mathbf{1}\mathbf{1}^\top/(1 - dh)$, and use it to recompute the group term $\hat{\boldsymbol{\epsilon}}_D^\top(\mathbf{I} - \mathbf{H}_D)^{-1}\hat{\boldsymbol{\epsilon}}_D$ in the proof of [Proposition 20.5.2\(e\)](#) from $\hat{\boldsymbol{\epsilon}}_D = \delta\mathbf{1}/(1 + mh_\star)$.

Exercise. B2

In simple regression with clean mean $\bar{x}_0$ and $S_0 = \sum (x_j - \bar{x}_0)^2$, show that $h_{j*} = 1/n_0 + (x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0$. In [Example \(Four identical bad records\)](#), with x_* far to the right and $\delta < 0$, which clean residuals does [Proposition 20.5.2\(e\)](#) push up, and which down? Explain why case 30 was flagged.

Solution. Use the centred rows $(1, x - \bar{x}_0)$, which span the same column space; then $\mathbf{A}^{-1} = \text{diag}(1/n_0, 1/S_0)$, so $h_{j*} = 1/n_0 + (x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0$. The shift in clean residual j is $-m\delta h_{j*}/(1 + mh_*)$, which has the sign of h_{j*} because $\delta < 0$. Since $x_* - \bar{x}_0 > 0$, h_{j*} grows with x_j : positive on the right of the clean data and negative on the far left (once $(x_j - \bar{x}_0)(x_* - \bar{x}_0)/S_0 < -1/n_0$). Residuals on the right are pushed up and those on the far left down. Case 30 is the clean case with the largest x , so it has the largest upward shift, and its studentized residual crosses two.

Exercise. B3

In the setting of [Proposition 20.5.2](#), delete $k < m$ of the cluster cases. Show that each deleted case has prediction residual $\delta/(1 + (m - k)h_*)$. What does this say about searching for a cluster of unknown size by deleting sets of size k ?

20.5.4.3 C. Going deeper**Exercise. C1**

Place two clusters of m identical bad cases at the two ends of a straight-line design, one above the clean line and one below. Describe, using [Theorem 20.5.1](#), what single-case deletion, deletion of one cluster, and deletion of both clusters report. Is it possible for each cluster to mask the other?

Exercise. C2

For a set D of cases, define $D_D = (\hat{\beta}_{(D)} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta}_{(D)} - \hat{\beta}) / (ps^2)$. Using [Theorem 20.5.1\(b\)](#), show that $D_D = \hat{\mathbf{e}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \mathbf{H}_D (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\mathbf{e}}_D / (ps^2)$.